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Healthy First-Degree Relatives From Multiplex Families vs Simplex Families Have Higher Subclinical Intestinal Inflammation, a Distinct Fecal Microbial Signature, and Harbor a Higher Risk of Developing Crohn's Disease

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BACKGROUND & AIMS: Unaffected first-degree relatives (FDRs) from families with ≥2 affected FDRs with Crohn's disease (CD, multiplex families) have a high risk of developing CD, although the underlying mechanisms driving this risk are poorly understood. We aimed to identify differences in biomarkers between FDRs from multiplex vs simplex families and investigate the risk of future CD onset accounting for potential confounders. **METHODS:** We assessed the Crohn's and Colitis Canada Genetic Environmental Microbial cohort of healthy FDRs of patients with CD. Genome-wide CD-polygenic risk scores, urinary fractional excretion of lactulose-to-mannitol ratio, fecal calprotectin (FCP), and fecal 16S ribosomal RNA microbiome were measured at recruitment. Associations between CD multiplex status and baseline biomarkers were determined using generalized estimating equations models. Cox models were used to assess the risk of future CD onset. **RESULTS:** There were 4051 participants from simplex families and 334 from CD multiplex families. CD multiplex status was significantly associated with higher baseline FCP ($P = .026$) but not with baseline CD-polygenic risk scores or the lactulose-to-mannitol ratio. Three bacterial genera were found to be differentially abundant between both groups. CD multiplex status at recruitment was independently associated with an increased risk of developing CD (adjusted hazard ratio, 3.65; 95% confidence interval, 2.18–6.11, $P < .001$). **CONCLUSION:** Within FDRs of patients with CD, participants from multiplex families had a 3-fold increased risk of CD onset, a higher FCP, and an altered bacterial composition, but not genetic burden or altered gut permeability. These results suggest that putative environmental factors might be enriched in FDRs from multiplex families.

Keywords: Crohn's Disease; Family History; Fecal Microbiome; Gut Inflammation; Epidemiology.

The familial nature of inflammatory bowel disease (IBD) has long been identified.[1,2] Observational studies have highlighted a positive family history of IBD as a major risk factor for developing the disease,[3,4] with this risk being particularly high in first-degree relatives (FDRs) of patients with Crohn's disease (CD). Importantly, there appears to be an additive risk increment of CD onset per additional affected FDR.[5] Therefore, individuals within families with multiple affected members, also known as multiplex families, harbor a notably high risk of developing CD. It has been shown that healthy individuals from multiplex families have a higher number of risk alleles in IBD-associated genes compared with healthy individuals from the general population and that the mean number of risk alleles increases with increasing numbers of affected FDRs.[5] However, none of the genetic markers could discriminate between individuals from single-case families (so-called simplex families) and those from multiplex families. Thus, genetics is unlikely the sole factor that can explain this increased risk,[5] raising the question of what could be driving the increased risk of CD in multiplex families.

In the present study, we aimed to identify differences between FDRs from multiplex families vs FDRs from simplex families regarding CD-related genetic risk, gut barrier dysfunction, preclinical gut inflammation, and fecal microbiota composition. We further aimed to investigate the risk of future development of CD in participants from multiplex families compared with participants from simplex families, accounting for potential confounders.

WHAT YOU NEED TO KNOW

BACKGROUND AND CONTEXT

First-degree relatives of patients with Crohn's disease from multiplex families have a notably high risk of developing Crohn's disease. The underlying mechanisms driving this risk are poorly understood.

NEW FINDINGS

Crohn's disease multiplex status is associated with higher subclinical gut inflammation and distinct fecal microbiome signatures, but not with genetic burden or gut barrier dysfunction. Crohn's disease multiplex status is associated with a 3-fold increased risk of developing Crohn's disease.

LIMITATIONS

Biomarkers and multiplex status were assessed only at recruitment, and most of the participants in the Crohn's disease multiplex families had only 2 affected first-degree relatives. Possibly rare allele variants with large effect sizes would not have been captured by the polygenic risk score.

CLINICAL RESEARCH RELEVANCE

This study confirms previous findings of population-based studies of an increased risk of Crohn's disease onset in first-degree relatives from multiplex families. We found that this risk is independent of subclinical gut inflammation, gut barrier dysfunction, and genetic burden. We hypothesize that putative environmental factors driving this risk might be enriched in first-degree relatives from multiplex families.

BASIC RESEARCH RELEVANCE

This study suggests that there are fecal microbiome signature differences between first-degree relatives from multiplex and simplex families. Future studies using novel exposome technologies may better capture the shared environmental factors contributing to a higher risk of disease in multiplex families.

Materials and Methods

Participant Recruitment

Participants were recruited as part of the Crohn's and Colitis Canada Genetic Environmental Microbial (CCC-GEM) project. The CCC-GEM project is an international, multicenter, prospective study recruiting healthy FDRs of patients with CD. The aim of the CCC-GEM project is to identify potential triggers associated with the later development of CD based on baseline characteristics and recruitment samples (serum, urine, and stool). The CCC-GEM recruited asymptomatic FDRs (siblings or offspring) of patients with CD aged between 6 and 35 years. Participants were screened using a standardized questionnaire to exclude individuals with the presence of any gastrointestinal symptoms, known diagnoses, or symptoms suggestive of IBD, celiac disease, or irritable bowel syndrome. Participants were also excluded if they had received antibiotic treatment within 3 months before recruitment. Demographic information and environmental risk data were recorded as previously described.[6]

After recruitment, participants were contacted by telephone every 6 months to assess for the development of new symptoms or new CD diagnosis. If the participants declared a change in health, including new symptoms or actual diagnosis, then we confirmed with their treating physician the diagnosis based on clinical, endoscopic, histologic, radiographic reports, or a combination of these. All participants or their guardians gave written informed consent. The study was approved by the Mount Sinai Hospital Research Ethics Board (Toronto–Global Project Office) and each of the local recruitment centers.

Study Population

This study included participants recruited in the CCC-GEM cohort between 2008 and 2017. For this analysis we included

*** Authors share co-first authorship; § Authors share co-senior authorship.**

Abbreviations used in this paper: **CCC-GEM, Crohn's and Colitis Canada Genetic Environmental Microbial; CD, Crohn's disease; CI, confidence interval; FCP, fecal calprotectin; FDRs, first-degree relatives; HR, hazard ratio; IBD, inflammatory bowel disease; IQR, interquartile range; IRR, incidence rate ratio; LMR, lactulose-to-mannitol ratio; OR, odds ratio; PRS, polygenic risk scores; rRNA, ribosomal RNA; SD, standard deviation; UC, ulcerative colitis.**

Most current article

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participants with an available demographic questionnaire at recruitment as of February 2021. Baseline demographics, detailed family history, race/ethnicity, family size, number of affected FDRs for each participant, and the relationship to the proband were collected.

We defined CD multiplex participants as those with ≥2 FDRs with CD at recruitment. We defined simplex participants as those with only 1 FDR with CD. In the primary analysis, we excluded participants with 1 FDR with CD and ≥1 FDRs with ulcerative colitis (UC) because we deemed these individuals not classifiable as CD multiplex. Participants with ≥2 FDRs with CD or UC were classified as IBD multiplex (see *Sensitivity Analyses*). We also excluded participants with UC diagnosed during follow-up and participants with an unconfirmed diagnosis of CD at the end of the study period. Participants with available baseline fecal calprotectin (FCP), urinary fractional excretion of lactulose-to-mannitol ratio (LMR), CD polygenic risk score (CD-PRS), or fecal 16S ribosomal RNA (rRNA) microbiome data were included.

Assessment of Biological Samples at Recruitment

Assessment of genetic risk score, gut barrier function, and subclinical intestinal inflammation. Assessment of the CD-related genetic risk was performed based on genotyping data available. Blood DNA was extracted using the Gentra Puregene Blood Kit (Qiagen; catalog #158389) as previously described.[7] Single-nucleotide polymorphism genotyping was then performed using the Immuno Beadchip (Illumina, Inc, San Diego, CA), HumanCoreExome-12 1.1 chip (Illumina, Inc), HumanCoreExome-24 1.0 chip (Illumina, Inc) or Illumina Human Global Screen Array assay (Illumina, Inc). Quality control and genotype imputation were conducted using standard criteria with the Michigan Imputation Server and the Haplotype Reference Consortium r1.1 panel.[8,9] The genome-wide CD-PRS were calculated using the LDpred 1.0.6 algorithm ($\rho = 0.1$),[10] which showed the best predictive ability to classify patients with IBD from healthy controls.[11] Based on summary statistics from a large-scale meta-analysis of European patients with IBD,[12] LDpred was run with default settings separately for each chromosome to reduce its computational burden with linkage disequilibrium reference genotypes corresponding to the imputed genotypes in the CCC-GEM cohort.[7]

Assessment of gut barrier function was performed using the urinary fractional excretion LMR from samples collected at recruitment, as previously described.[13–16]

Assessment of subclinical intestinal inflammation was performed using FCP from samples collected at recruitment, as previously described for the CCC-GEM cohort.[14]

Profiling of the fecal microbiota. Stool bacterial DNA extraction was performed using the QIAamp DNA Stool Mini Kit (Qiagen, Hilden, Germany). The V4 hypervariable region of bacterial 16S rRNA gene was amplified using the 515F/806R primer pair and sequenced in paired-end mode (2 × 150 base pairs) using the MiSeq platform (Illumina Inc).[17] Assembled reads were demultiplexed and processed by the quantitative insights into microbial ecology (QIIME2 2.0) software using the DADA2 pipeline using the default parameters.[18] QIIME2's q2-feature-classifier plugin against Silva database 132 of bacterial 16S rRNA gene sequences was used for taxonomic classification of marker-gene amplicon sequences.[19]

Statistical Analysis

For descriptive analysis, categorical variables are expressed as numbers and percentages, and continuous variables are described as median and interquartile range (IQR) or mean and standard deviation (SD). We used χ^2 to compare categorical variables and Wilcoxon's rank sum test to compare continuous variables between 2 groups. Association between CD multiplex status and baseline biomarkers (CD-PRS, LMR, and FCP) was performed using generalized estimating equations logistic and linear models, accounting for variability within family clusters. All generalized estimating equations models used an "exchangeable" correlation structure. For categorical variables, we estimated odds ratios (ORs) and their 95% confidence intervals (CIs) adjusted for age, sex, race/ethnicity, and family size. CD-PRS was assessed as a continuous variable. LMR and FCP were assessed both as continuous and categorical variables. LMR was log-transformed for analysis as a continuous variable. LMR was dichotomized with a threshold of 0.025 to define an abnormal gut barrier function.[16,20–22] FCP was dichotomized with a threshold of 100 μg/g to define the presence or absence of gut inflammation.[23–26]

Alpha diversity was determined on aggregated microbial amplicon sequence variants with identical genus taxonomic classification. We then generated a rarefaction curve (Supplementary Figure 1), and based on the results, rarefaction at a depth of 10,000 reads per sample was used, and samples with <10,000 reads were excluded. Rarefaction was performed using the phyloseq (v1.40.0) R package (R Foundation for Statistical Computing) and plotted using the rarecurve function from the vegan (v2.6-2) R package. Alpha diversity of the fecal microbial community was calculated using Shannon's diversity index at the genus level, and Wilcoxon's rank sum test was used to compare multiplex and simplex groups.

For beta diversity analysis, after the removal of singletons, principal coordinate analysis was performed based on the Bray-Curtis dissimilarity index at the genus level. Permutation analysis of variance was applied using the adonis function on distance matrices with 1000 permutations (vegan 2.6-2, R package). For the differential abundance analysis, singletons were removed, and the microbial amplicon sequence variants were aggregated at the genus level. Data normalization was done by converting the nonrarefied count table to relative abundance. Multivariate association with linear models (MaAsLin2 1.10.0) was used to ascertain associations between CD multiplex status and individual taxa, adjusting for potential confounders and accounting for variability within family clusters.[27] The Benjamini-Hochberg false discovery rate of $q < .05$ was considered significant after correction for multiple testing.

Finally, time to CD diagnosis between simplex and CD multiplex participants was analyzed by Kaplan-Meier curves, and a log-rank test was performed for group comparison. To account for clustering within families, marginal Cox models with robust variance estimates were used to assess the time-related risk of the future onset of CD based on simplex or CD multiplex status at recruitment.[28]

Participants who were lost during follow-up were right censored as being healthy at their latest visit. Also, if a second FDR from simplex families within the cohort developed CD or UC during the follow-up, individuals from the same family cluster were right-censored at the time of the second FDR's IBD

diagnosis. The hazard ratio (HR) with 95% CI was adjusted by demographics (age at recruitment, sex, race/ethnicity, and family size), FCP (continuous), LMR (continuous log-transformed), and CD-PRS, individually and all covariates combined. We assumed that missing data were missing at random. To address missing data, we used a complete case analysis approach using list-wise deletion, where any observation with missing values on any item used in the models was excluded from the analysis. We considered 2-sided P values $<.05$ as statistically significant. All analysis was performed R 4.2.1 software.

Sensitivity Analyses

Several sensitivity analyses were conducted to ensure the robustness of our findings. We assessed differences in biomarkers (CD-PRS, LMR, and FCP) and the time to CD onset by including participants with 1 FDR with CD and $\geq$1 FDRs with UC, defining this cohort as IBD multiplex. Furthermore, to evaluate the impact of subclinical gut inflammation on the fecal microbiome composition, we adjusted our analysis for demographics and FCP levels.

Additionally, we evaluated differences in the time to CD onset by including participants who were diagnosed with UC or had a suspected but unconfirmed diagnosis of CD during follow-up. We also analyzed the time to UC diagnosis and the overall time to IBD diagnosis (ie, CD or UC) between subjects from CD multiplex and simplex families.

Moreover, we considered the potential influence of undiagnosed CD at the time of recruitment by excluding individuals with follow-up periods of <1 year, 3 years, and 5 years. To assess the effect of the proband's age at CD onset on differences in biomarkers and the time to CD onset, we included this variable as a covariate in our models. To assess the impact of differentially abundant taxa between simplex and CD multiplex families in the future onset of CD, we conducted a post hoc sensitivity analysis using a Cox model, which included the taxa significantly associated at a false discovery rate significance level and demographics.

Results

Baseline Characteristics of the Cohort

The study included 4385 participants from the CCC-GEM cohort. The flow chart of the study is shown in Supplementary Figure 2. The median age at recruitment of the entire cohort was 17.0 years (IQR, 12.0-24.0 years) and 52.9% were female. The cohort comprised 4051 participants (92.4%) from simplex families and 334 participants (7.6%) from CD multiplex families. Participants from CD multiplex families had larger family sizes (mean, 5.7 [SD, 2.3] members vs 5.1 [SD 1.7] members, $P < .001$) and younger age of recruitment (median, 16.0 years [IQR, 11.0–22.0 years] vs 17.0 years [IQR, 12.0–24.0 years], $P = .022$). Relation to the proband was sibling in 67.8% (2745 of 4051) and offspring in 32.2% (1306 of 4051) in participants from simplex families. Among the 334 participants from CD multiplex families, 271 (81.1%) had a combined relation to the probands (sibling and offspring), whereas 56 (16.8%) had only affected siblings ($\geq$2), and 7 (2.1%) had both parents affected with CD. The proportion of Jewish Ashkenazi ethnicity was higher in participants from CD multiplex families (19.2% vs 9.2%, $P < .001$). Comparing participants from simplex vs CD multiplex families, there were no differences in sex, smoking status, country of recruitment, or follow-up time. Within participants from CD multiplex families, 295 participants had 2 FDRs (88.3%), 37 (11.1%) had 3 FDRs, and 2 (0.6%) had $\geq$4 FDRs affected with CD. Baseline characteristics are summarized in Table 1.

CD Multiplex Status Is Not Associated With Genetic Risk Burden Compared With Simplex

We found no significant differences comparing CD-PRS between those participants from CD multiplex and simplex families (Figure 1*A*). In the multivariable regression model, CD multiplex family status was not associated with higher CD-PRS after adjusting for demographic factors of age at recruitment, sex, race/ethnicity, and family size (estimate, 0.09; 95% CI −0.04 to 0.23; $P = .16$) (Table 2 and Supplementary Table 1).

CD Multiplex Status Is Not Associated With Altered Gut Barrier Dysfunction Compared With Simplex

We found no significant differences in LMR at recruitment between participants from CD multiplex and simplex families (Figure 1*B*). In the multivariable regression model, CD multiplex family status was not associated with increased baseline LMR after adjusting for age at recruitment, sex, race/ethnicity, and family size when assessed as a continuous variable (estimate −0.02; 95% CI, −0.04 to 0.01; $P = .14$) (Table 2) or as categorical variable with a cutoff of 0.025 (OR, 0.98; 95% CI, 0.76–1.27; $P = .91$) (Table 3). However, age at recruitment, male sex, and family size were independently associated with baseline LMR (Supplementary Table 2).

CD Multiplex Status Is Associated With Higher Subclinical Intestinal Inflammation Compared With Simplex

FCP at recruitment was significantly higher in participants from CD multiplex families compared with those from simplex families (61.1 μg/g [IQR, 31.7–109.8 μg/g] vs 50.2 μg/g [IQR, 30.1–85.3 μg/g]) (Figure 1*C*). In the multivariable regression model, when adjusting for demographics (age at recruitment, sex, race/ethnicity, and family size), CD multiplex status compared with simplex remained significantly associated with higher baseline FCP levels (estimate, 37.74; 95% CI, 4.41–71.10; $P = .026$) (Table 2 and Supplementary Table 3). This association was also significant when assessing FCP as categorical variable with a cutoff of 100 μg/g (OR, 1.51; 95%, CI 1.20–1.91; $P = .004$) (Table 3).

CD Multiplex Status Is Associated With Differences in Fecal Microbial Composition Compared With Simplex

We found no significant differences in alpha ($P = .50$) or beta (permutation analysis of variance: $R^2 = 2^{-04}$, $P = .64$) diversity between participants from CD multiplex and simplex families (Figure 2*A* and *B*). However, after adjusting for demographics (age at recruitment, sex, race/ethnicity, and family

Table 1. Baseline Characteristics of the Study Population

Variable	Simplex (n = 4051)	CD multiplex (n = 334)	*P* value[a]
Age at recruitment, *y*	17.0 (12.0-24.0)	16.0 (11.0-22.0)	.022
Male sex	1910 (47.1)	155 (46.4)	.84
Smoker at recruitment	208 (5.2)	14 (4.2)	.51
Country of recruitment			.09
Canada	2216 (54.7)	191 (57.2)	
United States of America	531 (13.1)	40 (12.0)	
Israel	648 (16.0)	64 (19.2)	
Other	656 (16.2)	39 (11.7)	
Family size, mean (SD), n[b]	5.1 (1.7)	5.7 (2.3)	<.001
Relation to proband			<.001
Sibling	2745 (67.8)	56 (16.8)	
Offspring	1306 (32.2)	7 (2.1)	
Sibling + offspring	0 (0.0)	271 (81.1)	
Race/ethnicity			<.001
Asian	138 (3.4)	3 (0.9)	
Black	52 (1.3)	2 (0.6)	
Jewish-Ashkenazi	373 (9.2)	64 (19.2)	
Jewish–other	307 (7.6)	21 (6.3)	
White–non-Jewish	2917 (72.0)	222 (66.5)	
Other	264 (6.5)	22 (6.6)	
Follow-up, y	5.6 (3.4-8.6)	5.7 (3.2-8.9)	.83
Total affected FDRs			
1	4051 (100)		
2		295 (88.3)	
3		37 (11.1)	
4		2 (0.6)	

NOTE. Categorical variables are expressed as n (%), and continuous variables are expressed as median (IRQ) or as indicated otherwise as mean (SD).
[a]Comparison was made using the χ^2 test or Wilcoxon's test.
[b]Family size was defined as the sum of parents, siblings, and offspring within the family cluster.

size), 3 bacterial genera were found to be differentially abundant in participants from CD multiplex or simplex families after correcting for multiple testing ($q < .05$) (Figure 2*C* and Supplementary Table 4). The genus *Prevotella* was enriched in participants from CD multiplex families ($q = .026$), whereas the genera *Eisenbergiella* ($q = .006$) and *Dielma* ($q = .018$) were enriched in participants from simplex families. In sensitivity analysis, we further adjusted for demographics and FCP, and interestingly, the genus *Prevotella* and the genera *Eisenbergiella* and *Dielma* remained statistically associated with CD multiplex and simplex status, respectively, even after adjusting for subclinical intestinal inflammation (Supplementary Table 5).

CD Multiplex Status Is Associated With a Higher Risk of Future Onset of CD Independently of Genetic Risk, Gut Barrier Function, and Subclinical Inflammation

Follow-up was available for 4211 participants in the cohort, and after a median of 5.56 years (IQR, 3.40–8.60 years), 81 participants were diagnosed with CD, and 4130 remained asymptomatic. There were no differences in follow-up duration between participants from CD multiplex families vs simplex families (median, 5.67 years [IQR. 3.24–8.86 years] vs 5.60 years [IQR, 3.40–8.60 years], respectively; $P = .83$). Within participants who developed CD, the median time from recruitment to diagnosis was 3.05 years (IQR, 1.37–4.65 years).

We found a significantly higher proportion of CD multiplex participants at recruitment in the group who later developed CD vs those who remained asymptomatic at the last follow-up (23.45% [95% CI, 15.10%–34.40%] vs 7.32% [95% CI, 6.57%–8.15%], $P < .001$). When taking the time to event into account, the risk of CD onset was significantly higher in participants from CD multiplex families compared with participants from simplex families (log-rank test, $P < .001$) (Figure 3).

To adjust for potential confounding factors on CD onset, we performed a multivariable Cox model adjusted for age at recruitment, sex, race/ethnicity, and family size. CD multiplex family status at recruitment was independently

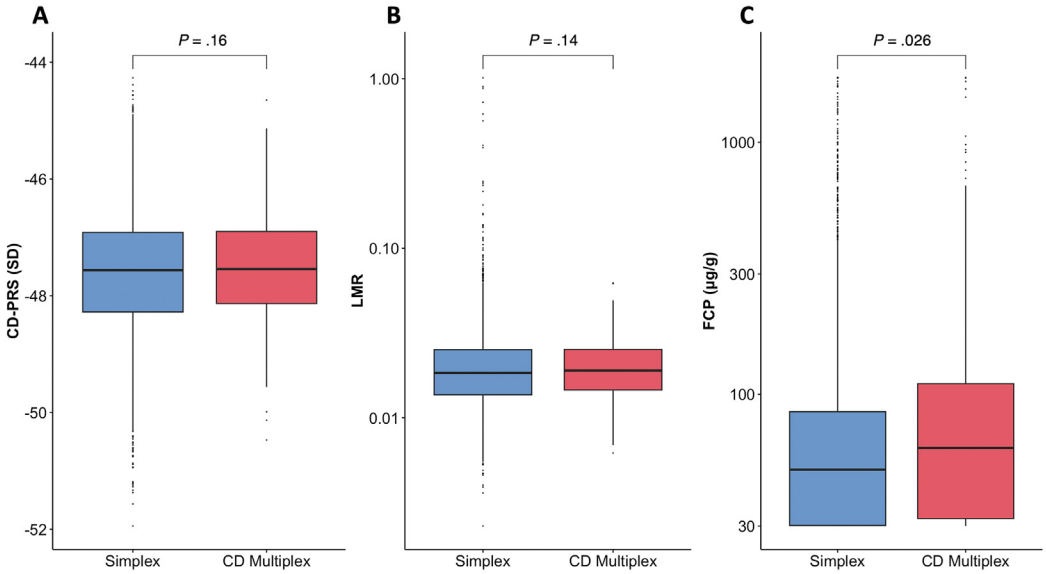


Figure 1. Biomarkers at recruitment. (*A*) CD-related genetic burden assessed by CD-PRS. (*B*) Gut barrier permeability at recruitment, measured by urinary fractional LMR. (*C*) Subclinical inflammation at recruitment, measured by FCP. *Box plot*: The *horizontal line* in the middle of each box indicates the median; the *top and bottom borders* of the box mark the 75th and 25th percentiles, respectively, and the *vertical lines* denote the lowest and highest values within 1.5 IQR from the first and third quartiles, respectively. *P* values for comparisons between groups are based on generalized estimating equations adjusted by age at recruitment, sex, race/ethnicity, and family size. CD-PRS data were available for 2990 individuals. Baseline LMR determination was available for 3250 individuals. Baseline FCP determination was available for 3623 individuals.

associated with an increased risk of developing CD after adjusting for demographics (HR, 3.65; 95% CI, 2.18–6.11; $P < .001$). We also developed models adjusting for demographics and adding FCP, LMR, and CD-PRS individually and all variables combined (Table 4). Notably, in all the models, CD multiplex family status was independently associated with an increased risk of CD onset with a similar effect size. When all variables were combined (demographics, FCP, LMR, and CD-PRS), CD multiplex status remained statistically associated with disease onset (adjusted HR, 3.46; 95% CI, 1.72–6.97) (Supplementary Table 6).

To mitigate the potential influence of undiagnosed CD at the time of recruitment, we conducted sensitivity analyses wherein we excluded participants with follow-up durations of <1, 3, and 5 years. CD multiplex status at recruitment remained independently associated with a FCP >100 µg/g, even when excluding participants with <1, 3, and 5 years of follow-up (Supplementary Tables 7, 8, and 9). Importantly, CD multiplex status at recruitment remained associated

Table 2. Generalized Estimating Equations Models for Different Biomarkers

Variable (continuous)	Estimate	Standard error	95% CI	*P* value
CD-PRS (LDpred)	0.09	0.07	−0.04 to 0.23	.16
LMR (log_{10})	−0.02	0.01	−0.04 to 0.01	.14
FCP	37.74	17.01	4.41 to 71.10	.026

NOTE. Outcomes are presented as continuous variables. Each row is a separate analysis. The estimates, standard errors, 95% CI, and *P* values are for the CD multiplex (vs simplex) variable on each of the outcomes. All models adjusted were for demographic factors (ie, age at recruitment, sex, race/ethnicity, and family size) and accounted for family clusters.
CD-PRS data were available for 2990 individuals (68.2%), of which 221 (7.4%) were CD multiplex. Baseline LMR determination was available for 3250 individuals (74.1%), of which 245 (7.5%) were CD multiplex. Baseline FCP determination was available for 3623 individuals (82.6%), of which 277 (7.7%) were CD multiplex.

Table 3. Generalized Estimating Equations Models for Different Biomarkers

Variable (categorical)	OR	95% CI	*P* value
LMR 0.025	0.98	0.76–1.27	.91
FCP 100	1.51	1.20–1.91	.004

NOTE. Outcomes as categorical variables. LMR 0.025 indicates a threshold of 0.025 to define an abnormal gut barrier function. FCP 100 indicates FCP cutoff at 100 µg/g to define the presence or absence of gut inflammation. Each row is a separate analysis. The OR, 95% CIs, and *P* values are for the CD multiplex (vs simplex) variable on each of the outcomes. All models adjusted were for demographic factors (ie, age at recruitment, sex, race/ethnicity, and family size) and accounted for family clusters. Baseline LMR determination was available for 3250 individuals (74.1%), of which 245 (7.5%) were CD multiplex. Baseline FCP determination was available for 3623 individuals (82.6%), of which 277 (7.7%) were CD multiplex.

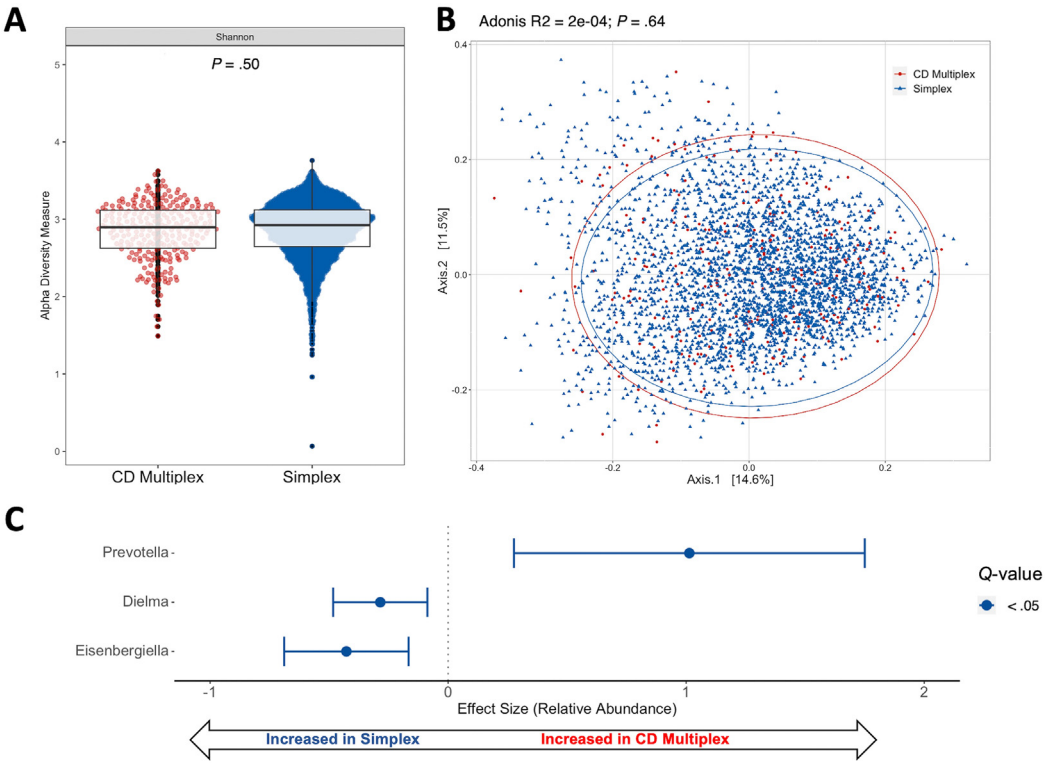


Figure 2. Baseline fecal sample for 16S rRNA gene microbiome analysis was available from 3421 participants. (*A*) Alpha diversity as assessed by the Shannon's diversity index. (*B*) Beta diversity expressed by the Bray-Curtis dissimilarity index at the genus level. The *ellipsoids* represent a 95% CI surrounding each group. (*C*) Coefficient plot shows the effect size and 95% CI (*x*-axis) for the 3 taxa at the genus level in fecal samples of participants associated ($q < .05$) with CD multiplex (positive effect size; *right to the dotted line*) and simplex (negative effect size; *left to the dotted line*) after adjusting for age at recruitment, sex, race/ethnicity, and family size.

with future onset of CD even after excluding subjects with less than 1, 3, and 5 years of follow-up in all the models (Supplementary Tables 10, 11, and 12).

We assessed differences in the time to CD onset, including participants diagnosed with UC or those with suspected but unconfirmed CD during follow-up. The results

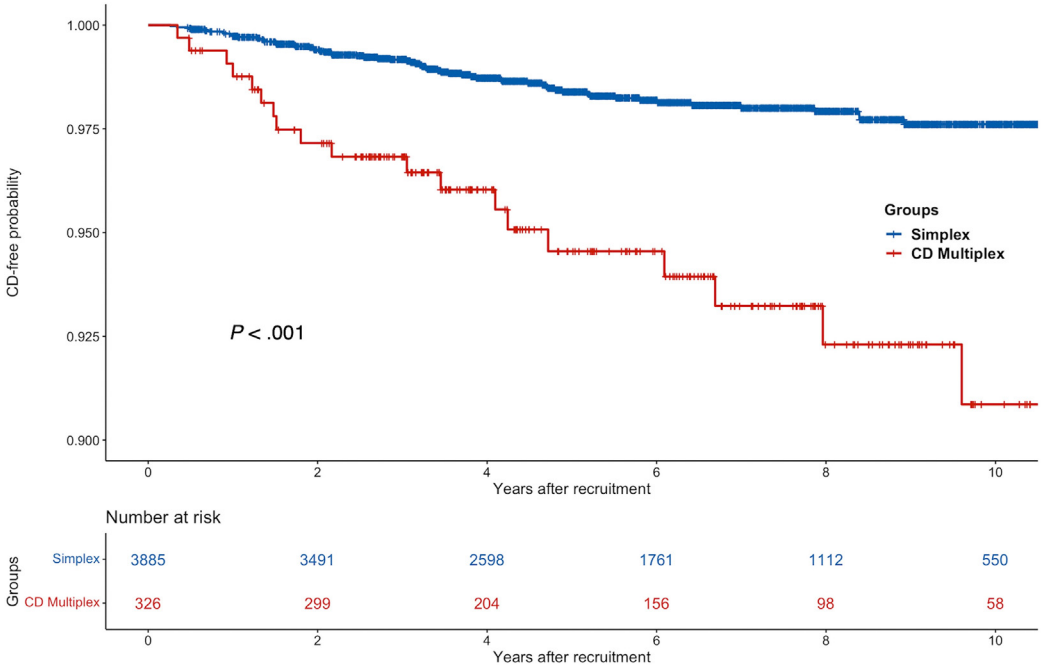


Figure 3. Kaplan-Meier analysis of time to CD onset during follow-up. The *P* value was based on the log-rank test.

Table 4. Cox Models for the Risk of Crohn's Disease Onset of Crohn's Disease Multiplex vs Simplex Status at Recruitment

Variables	HR	95% CI	*P* value
Demographics	3.65	2.18–6.11	<.001
Demographics + FCP	3.22	1.85–5.61	<.001
Demographics + LMR (log_{10})	3.84	2.11–6.98	<.001
Demographics + CD-PRS	3.57	2.02–6.30	<.001
Demographics + FCP + LMR (log_{10}) + CD-PRS	3.46	1.72–6.97	.001

NOTE. Demographics include age at recruitment, sex, race/ethnicity, and family size. Each row is a separate model. The HR, 95% CIs, and *P* values are for the CD multiplex (vs simplex) variable on each of the models. All models accounted for family clusters.

were consistent with our main analysis. In this sensitivity analysis, CD multiplex status was also associated with a higher risk of IBD onset (including both CD and UC), but not with UC alone, likely due to the small number of new UC diagnoses (n = 12) (Supplementary Table 13).

Including the proband's age at the time of their CD diagnosis in all the models showed consistent results with the main analyses (Supplementary Tables 14 and 15). Moreover, including the relative abundance of the genera *Prevotella, Eisenbergiella*, and *Dielma* in the time to CD onset analysis showed that CD multiplex status remained associated with an increased risk (Supplementary Table 16).

IBD Multiplex Is Associated With Higher Subclinical Intestinal Inflammation and Future Onset of CD

We assessed differences in biomarkers and the time to CD onset, including participants with 1 FDR with CD and ≥1 FDRs with UC in the multiplex group (defined as IBD multiplex). Consistent with our main analysis, IBD multiplex status at recruitment was associated with a higher FCP level (Supplementary Table 17) and a higher risk of CD onset (Supplementary Figure 3 and Supplementary Table 18).

Discussion

To the best of our knowledge, this is the largest prospective cohort study to date examining the effect of being a member of a CD multiplex family on the risk of disease onset and its relationship with several biological factors that potentially may be involved in this observed increased risk. Importantly, we found a 3-fold higher risk of future CD onset associated with the CD multiplex status at recruitment compared with FDRs from simplex families. This risk was independent from genetic burden, gut barrier function, and subclinical intestinal inflammation. This stability suggests that the familial aggregation of CD risk is not solely attributable to these known biological markers and genetic predispositions. This robust association implies that there are additional, yet unidentified, factors shared within multiplex families that contribute to the heightened risk of CD. We further identified 3 bacterial taxa to be differentially abundant when comparing individuals from simplex or multiplex families.

Population-based studies have shown that a positive family history of IBD is the strongest risk factor for IBD incidence. The effect size of this risk is notably higher in FDRs of patients with CD than in FDRs of patients with UC, second-degree, or third-degree relatives of patients with CD.[3,4] A Danish population-based study found that the risk of CD onset is approximately 8-fold higher in FDRs of individuals with CD compared with the general population (adjusted incidence rate ratio [IRR], 7.77; 95% CI, 7.05-8.56)[3] and was more pronounced in individuals with ≥2 FDRs with IBD (adjusted IRR, 9.77; 95% CI, 7.07–13.50).[3] A recent Korean population-based cohort study revealed an even higher risk of CD onset in individuals with an FDR with CD (IRR, 22.2; 95% CI, 20.5-24.5), which was markedly increased to 107.5-fold (IRR, 107.5; 95% CI, 57.84-199.8) in individuals with ≥2 FDRs affected.[4]

One limitation of population-based studies is the challenge of adjusting for known CD risk-related biomarkers. The intrinsic characteristics of the CCC-GEM cohort, with the availability of biomarkers at recruitment, allowed us to account for several of these factors involved in the risk of CD onset in the predisease stage. We showed no major effect of genetic burden between FDRs from CD multiplex and simplex families, which suggests that nongenetic differences may account for the increased risk in multiplex individuals. Joossens et al[5] investigated polymorphisms of specific genes (*NOD2*, *NOD1*, *TLR4*, and *CARD8*) among multiplex families. They found a significant correlation between the number of genetic mutations and an increasing number of FDRs with CD, supporting the hypothesis of a heavier genetic predisposition among multiplex families.

However, a recently published study investigated the known common variants associated with the risk of IBD in 8 multiplex families by weighted genetic risk score calculated by previous large a genome-wide association study.[29] They found only 1 multiplex family with a significantly higher genetic burden of common CD-risk variants. In our study, the absence of genetic differences between both groups could be explained for several reasons.

First, the PRS, originally described by Khera et al[11] was designed to identify individuals with risk equivalent to monogenic mutations of developing IBD within the general population. We, instead, assessed participants at risk of developing CD within a cohort of healthy FDRs with affected family members of CD (not the general population), where genetic differences could presumably be smaller.

Second, 88.3% of the multiplex participants had 2 FDRs affected with CD, 11.1% had 3 FDRs, and 0.60% had 4 FDRs. Hence, the PRS might not capture the differences among participants with 1 FDR affected with CD and participants with 2 affected FDRs.

Third, the lack of assessment of rare allele variants but with large effect sizes instead of known common IBD

variants might not be easily captured in summary statistics from a large meta genome-wide association study used for CD-PRS calculations. Thus, it remains possible that whole-genome sequencing of multiplex families might still reveal a genetic risk associated with multiplex families.

Increased intestinal permeability has been described in individuals with CD,[30] and in healthy FDRs, it is associated with a higher risk of CD onset as well as with altered gut microbiome composition and functions.[14,16] In contrast, we did not find a difference in intestinal permeability as measured by LMR between participants from CD multiplex and simplex families. Given that our control group was asymptomatic FDRs from simplex families and not the general population, differences might not be significant enough to be detectable. Another potential reason for the result is that intestinal permeability may be a consequence and not a primary phenomenon in CD onset among multiplex families.

Moreover, we investigated the association between subclinical intestinal inflammation and CD multiplex status. Previous studies have shown that asymptomatic FDRs of patients with CD harbor abnormal levels of FCP and a disease-associated gut microbiome.[31,32] Consistent with these studies, we found that CD multiplex family status is associated with a higher FCP and distinct fecal microbiome signatures within FDRs. One hypothesis could be that unaffected family members of multiplex families likely share factors with the CD probands that lead to dysregulated immune pathways, dysbiosis, and progression to subclinical inflammation during the preclinical phase, and in turn, this may confer increased susceptibility to the later development of CD. Indeed, healthy FDRs from multiplex families have an increased inflammatory state characterized by serum proteomic signatures that might be associated with future onset of CD.[33,34]

We also investigated the association between gut microbiota composition and CD multiplex status, because it has been previously linked with the future onset of CD.[35] To our knowledge, our study is the first to identify several bacterial taxa differentially abundant in FDRs from CD multiplex vs simplex families. These signatures might indicate that participants from CD multiplex families share some common fecal microbiome traits with established CD. Among them, the genus *Prevotella* was enriched in participants from CD multiplex families. This taxon has been linked to inflammation in immune-mediated inflammatory diseases (including IBD), especially through a T-helper type 17–mediated inflammation.[36–42] Sila et al[43] found a distinct fecal microbial community in newly diagnosed, treatment-naïve IBD pediatric patients compared with their healthy siblings and unrelated healthy controls, with higher levels of *Prevotella*.

Furthermore, a murine model study showed that intestinal colonization of mice with *Prevotella* leads to alteration of the gut microbial community, reduction of short-chain fatty acids, and exacerbation of intestinal inflammation.[44] Given that in a sensitivity analysis, we adjusted the association between CD multiplex and the relative abundance of *Prevotella* with subclinical intestinal inflammation, we might hypothesize that the increase in *Prevotella* might be an early change involved in the predisease phase of participants from multiplex families that facilitates an increased risk of CD onset. In contrast, we found the genus *Eisenbergiella* was depleted in individuals from CD multiplex families vs those from simplex families. A study found a lower relative abundance of *Eisenbergiella* in CD patients compared with healthy controls in ileal biopsy specimens.[45] Another study also reported lower levels of *Eisenbergiella* in fecal samples of patients with UC compared with healthy controls.[46]

Affected siblings in multiplex Ashkenazi Jewish families were more likely to be consecutively affected based on birth order and proximity of age, supporting that shared environment at a specific time point might contribute to disease risk.[47] Indeed, this is supported by a recent, population-based nested case-control study that identified several environmental risk factors for later onset of the disease among FDRs of patients with IBD.[48] In a recently published study, Xue et al[49] described several environmental factors associated with future onset of CD in healthy FDRs. A larger household in the first year of life and living with a canine pet between the ages of 5 and 15 years were found to be protective, whereas living with a bird pet at recruitment was associated with a higher risk of CD onset.[49] Our results suggest that the increased risk seen in participants from multiplex families may reflect the contribution of an altered shared environment.

Finally, after adjusting for demographics and several biomarkers, including genetic burden, subclinical inflammation, and gut barrier function, we demonstrated a 3-fold increased risk of disease onset associated with CD multiplex family status compared with simplex at recruitment. Strikingly, we found a consistent effect size when adjusting for each of these factors individually and combined. In the absence of an obvious genetic risk factor difference between participants from CD multiplex and simplex families, a deeper exploration of potential environmental differences between multiplex and simplex families is warranted.

This study has several strengths. This cohort is the largest to date in which healthy participants from CD multiplex families with a median follow-up of >5 years were prospectively assessed. Moreover, we assessed the association of CD multiplex with several important biomarkers, such as the fecal microbiome, FCP, CD-PRS, and gut barrier function, which are all possibly involved in CD pathogenesis and were previously shown to confer risk of future onset of CD. Several sensitivity analyses showed the robustness of our results.

Given that CD usually has a long preclinical phase, it might be argued that the association between CD multiplex status and higher subclinical gut inflammation may be due to undiagnosed disease. A recent population-based case-control study from Denmark found differences in biochemical and hematologic parameters up to 8 years before diagnosis in patients with CD vs healthy controls.[50] Importantly, the associations we report of increased risk of CD onset and higher subclinical gut inflammation between participants from CD multiplex and simplex families remained significant, even when excluding individuals with <5 years of follow-up.

Notwithstanding, this study has some limitations. First, we did not include some factors that might contribute to disease onset, such as environmental factors. Questionnaire-based assessments of environmental factors are limited to a set of questions that cannot measure unknown or novel environmental exposures and are prone to recall bias. With novel exposome technologies, future studies may better capture the shared environmental factors contributing to a higher risk of disease in multiplex families. Moreover, household sharing at the time of recruitment could have provided valuable insights into the impact of environmental factors on biomarker differences and future risk of CD. Unfortunately, we did not have information on household sharing between family members at the time of recruitment.

Second, each measured biomarker may have intrinsic limitations related to the technique or approach; notably, LMR is an indirect measure of intestinal paracellular permeability and, therefore, may not reflect specific alterations in tight junction proteins. Additionally, newer, more precise assays to assess gut permeability[51] were not available at the time of this study and should be considered for use in future studies.

Third, the biomarker values and multiplex status were only defined at the time of recruitment, and thus, we could not assess the dynamic changes of these variables over time.

Fourth, most of the participants in the CD multiplex families had only 2 affected FDRs vs 1 FDR affected in the simplex group. It is possible that we were not able to capture subtle differences in biomarkers, notably the genetic burden.

Fifth, the sample size of the subgroup of CD multiplex with $\geq$3 affected FDRs was too small and did not allow appropriate analyses to evaluate whether there was a dose response to the observations we reported.

Finally, it is important to note that most of patients with CD do not have an FDR with CD, and therefore, the results described in this study cannot be generalized to all patients with CD.

Conclusion

In conclusion, we showed that FDRs from multiplex families have a 3-fold higher risk of CD onset, even after adjusting for demographics, subclinical inflammation, gut barrier function, and genetic burden compared with individuals from simplex families. Furthermore, we found that high FCP was associated with CD multiplex as well as differentially abundant fecal bacterial taxa. We did not find associations between CD multiplex status and gut barrier function or genetic risk. Future studies should comprehensively assess environmental conditions (ie, dietary patterns, metabolomics, and early-life exposomes) to fully unravel the factors driving this increased risk of disease in FDRs from multiplex families potentially involved in the risk of developing the disease in CD multiplex families.

Supplementary Material

Note: To access the supplementary material accompanying this article, visit the online version of *Gastroenterology* at www.gastrojournal.org, and at https://dx.doi.org/10.1053/j.gastro.2024.08.031.

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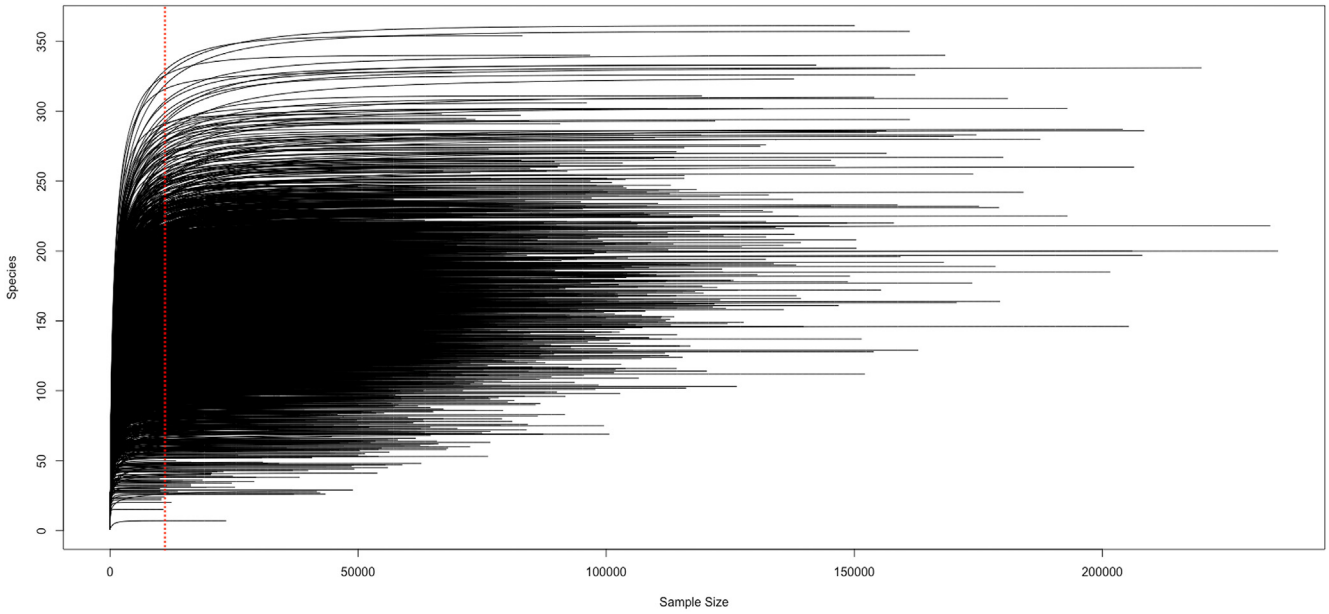
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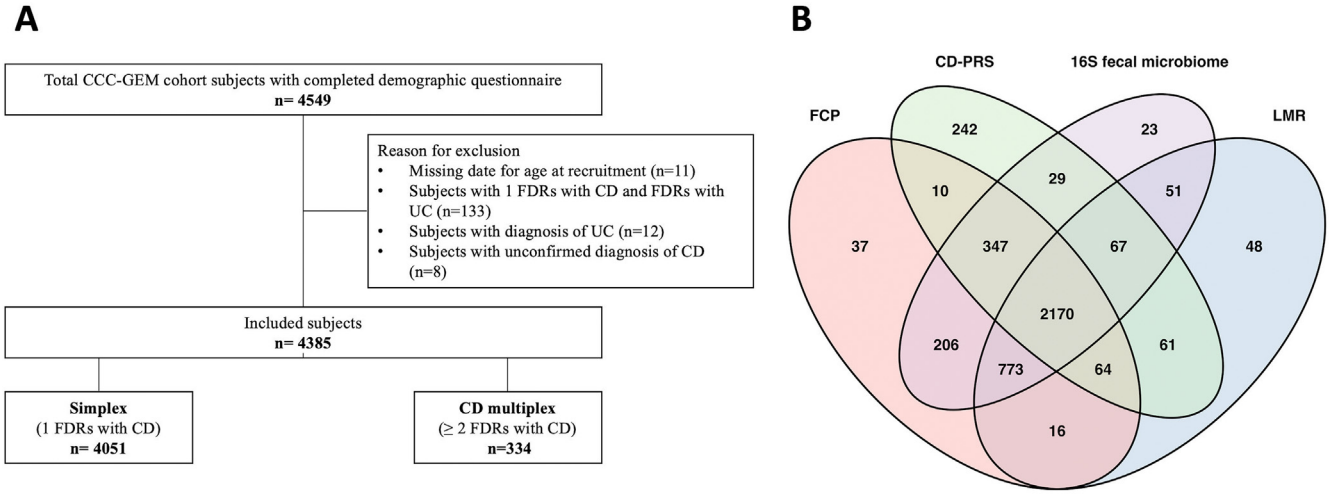
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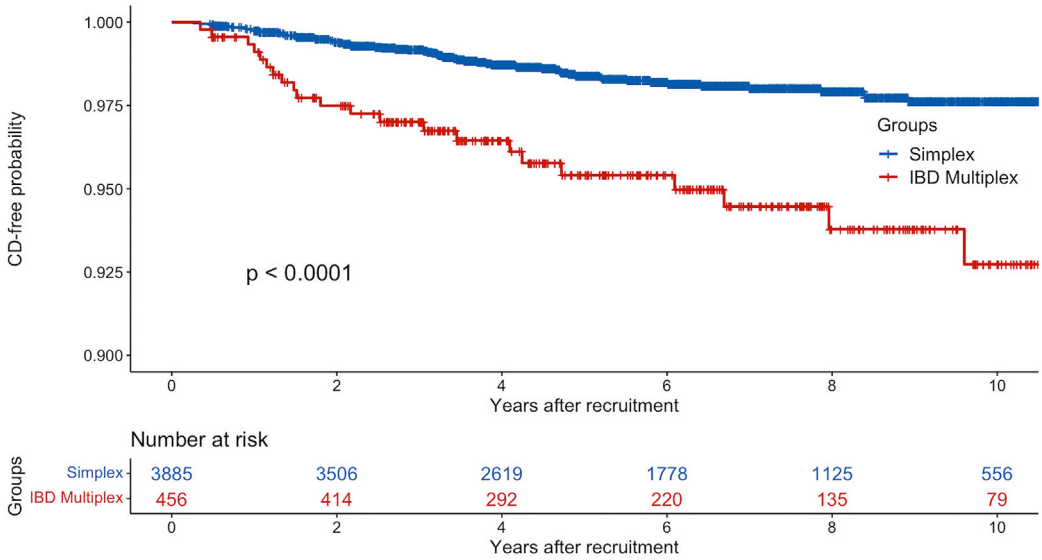
Data Availability
Requests for additional raw and analyzed data should follow the instructions given at http://www.gemproject.ca/data-access/. All submissions will be reviewed by the Genetic Environmental Microbial (GEM) Project Operating Committee to ensure that the requested samples/data will not interfere in any way with the intended GEM Project analysis of the nested cohort as per the original GEM Project Study Design and is not a duplication of analysis already ongoing. Those proposals meeting this evaluation will be distributed to all members of the GEM Project Steering Committee for review and open discussion. This review will focus on the global scientific merit of the proposal and the availability of requested samples and data, ensuring there is no compromise of the original intent of the GEM project. It would be of value to contact a member of the Steering Committee who could help sponsor your application. Those projects achieving a majority vote of approval at the GEM Project Steering Committee will be informed that the GEM Project will provide a letter of support stating that the requested samples or data will be made available to the applicants once the applicant receives funding from a granting agency that applies an independent peer review process to the proposal. The criteria to be used for review of all submissions will include the "scientific relevance" of the proposal and the judged availability of biological material requested. The budget to be requested from a funding agency must allow for any expenses in processing samples or in setting up the appropriate queries of the database. The intent is to allow sufficient time for applicants to consider a submission for funding opportunities



Supplementary Figure 1. Rarefaction curves of 16S fecal samples in the GEM cohort with available environmental risk assessment (ERA) questionnaire data. Rarefaction plot shows all samples with >10,000 reads (n = 3420) that were sequenced to a depth enough to capture the richness present in each sample. The *x-axis* represents the sampling depth of the sequencing reads. The *y-axis* represents the amplicon sequence variants richness captured at a given sampling depth. The *vertical red dashed line* indicates 10,000 reads threshold.



Supplementary Figure 2. (*A*) Flow chart of the study. (*B*) Number of individuals with available biomarkers.



Supplementary Figure 3. Kaplan-Meier analysis of time to CD onset during follow-up of IBD multiplex vs simplex at recruitment.